Database & Management System

[LAB MANUAL]

K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B.TECH CSE (CORE)

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LAB SHEET-2

Objective:

- To understand and implement SQL transactions using-

  COMMIT, SAVEPOINT, and **ROLLBACK.**

- Retrieve records using **SELECT** queries, and perform **INSERT, UPDATE**, and **ALTER** operations on tables.

Instructions:

1. Open your preferred Database Management System (DBMS) tool.
2. Create the required tables: CLASS, EMPLOYEES, and Client_master.
3. Insert the given/suitable records into each table.
4. Execute the SQL scripts and queries step by step.
5. Observe the output carefully after each operation, especially after ROLLBACK and UPDATE commands.

Table Structures:

CLASS

D (NUMBER) and Name (VARCHAR).

The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

EMPLOYEES

Employee_Id (NUMBER), First_Name (VARCHAR), Last_Name (VARCHAR), Email (VARCHAR), Phone_Number (VARCHAR), Hire_Date (DATE), Job_Id (VARCHAR), Salary (NUMBER), Commission_Pct (NUMBER), Manager_Id (NUMBER), and Department_Id (NUMBER).

DDL Questions (Data Definition Language):

1. How do you create a table named CLASS with appropriate data types?
2. How do you create a table named EMPLOYEES with the specified attributes?
3. How do you create a table named Client_master with the specified attributes?
4. What SQL statement would you use to rename the table Client_master to Client12?
5. How do you display the bal_due column heading as “BALANCE”?

DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

- INSERT INTO class VALUES (5,’Rahul’);
- COMMIT;
- UPDATE class SET name = ‘Abhijeet’ WHERE id=’5’;
- SAVEPOINT A;
- INSERT INTO class VALUES (6,’Chris’);
- SAVEPOINT B;
- INSERT INTO class VALUES (7,’Bravo’);
- SAVEPOINT C;
- SELECT * FROM class;
- ROLLBACK TO B;
- SELECT * FROM class;
- ROLLBACK TO A;

Part B: Retrieving Records (EMPLOYEES Table)

1. Find out the employee id, names, and salaries of all the employees.
2. List out the employees who work under manager 100.
3. Find the names of the employees who have a salary greater than or equal to 4800.
4. List out the employees whose last name is ‘AUSTIN’.
5. Find the names of the employees who work in departments 60, 70, and 80.

6. Display the unique Manager_Id.

Part C: Updating Records (Client_master Table)

1. Insert five records into the Client_master table.
2. Find the names of clients whose bal_due > 5000.
3. Change the bal_due of ClientNO "C123" to Rs. 5100.

Create a **new schema (database)-**

```
CREATE DATABASE experiment4;
USE experiment4;
```

Create **Tables-**

```
CREATE TABLE CLASS (
   ID INT PRIMARY KEY,
   Name VARCHAR(50)
);
```

Create **Employee Table-**

```
CREATE TABLE EMPLOYEES (
   Employee_Id INT PRIMARY KEY,
   First_Name VARCHAR(50),
   Last_Name VARCHAR(50),
   Email VARCHAR(100),
   Phone_Number VARCHAR(20),
   Hire_Date DATE,
   Job_Id VARCHAR(20),
   Salary INT,
   Commission_Pct DECIMAL(5,2),
   Manager_Id INT,
```

```
    Department_Id INT
);
```

Create Client_master Table-

```
CREATE TABLE Client_master (
    ClientNO VARCHAR(10) PRIMARY KEY,
    Name VARCHAR(50),
    Address VARCHAR(100),
    Bal_due INT
);
```

Insert Initial Records-

```
INSERT INTO CLASS VALUES (1,'Bravo'), (2,'Alex'), (4,'Cheng');
```

EMPLOYEES sample records-

```
INSERT INTO EMPLOYEES VALUES
(101,'Amit','Sharma','amit@example.com','9999999999','2020-01-01','DEV',5000,0.10,100,60),
(102,'Neha','Verma','neha@example.com','8888888888','2021-02-15','HR',4800,NULL,101,70),
(103,'Ravi','Kumar','ravi@example.com','7777777777','2019-05-20','MGR',6000,0.15,100,80),
(104,'Priya','Singh','priya@example.com','6666666666','2022-07-10','DEV',4500,NULL,102,60);
```

Client_master sample records-

```
INSERT INTO Client_master VALUES
('C123','Ravi','Delhi',5000),
('C124','Neha','Mumbai',6000),
('C125','Amit','Chennai',4500),
('C126','Priya','Kolkata',7000),
```

('C127','Arjun','Pune',3000);

DDL Operations-

Rename Client_master to Client12-

RENAME TABLE Client_master TO Client12;

Display bal_due column as BALANCE-

SELECT Bal_due AS BALANCE FROM Client12;

11	21:19:15	CREATE TABLE CLASS (ID INT PRIMARY KEY, Name VARCHAR(50))	0 row(s) affected	0.171 sec
12	21:19:25	CREATE TABLE EMPLOYEES (Employee_Id INT PRIMARY KEY, First_Name VARCHAR(50), Last_Na...	0 row(s) affected	0.078 sec
13	21:19:35	CREATE TABLE Client_master (ClientNO VARCHAR(10) PRIMARY KEY, Name VARCHAR(50), Addres...	0 row(s) affected	0.046 sec
14	21:19:47	INSERT INTO CLASS VALUES (1,'Bravo'), (2,'Alex'), (4,'Cheng')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
15	21:19:56	INSERT INTO EMPLOYEES VALUES (101,'Amit','Sharma','amit@example.com','9999999999','2020-01-01','DEV...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
16	21:20:01	INSERT INTO Client_master VALUES ('C123','Ravi','Delhi',5000), ('C124','Neha','Mumbai',6000), ('C125','Amit','C...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
17	21:20:13	RENAME TABLE Client_master TO Client12	0 row(s) affected	0.047 sec
18	21:20:18	SELECT Bal_due AS BALANCE FROM Client12 LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

BALANCE
5000
6000
4500
7000
3000

Client12 1 × Read Only

Transactions and Savepoints (CLASS Table)-

SET autocommit = 0;

START TRANSACTION;

INSERT INTO CLASS VALUES (5,'Rahul');

COMMIT;

UPDATE CLASS SET Name = 'Abhijeet' WHERE ID = 5;

SAVEPOINT A;

INSERT INTO CLASS VALUES (6,'Chris');

SAVEPOINT B;

INSERT INTO CLASS VALUES (7,'Bravo');

SAVEPOINT C;

4	21:17:06	SET autocommit = 0	0 row(s) affected	0.000 sec
5	21:17:06	START TRANSACTION	0 row(s) affected	0.000 sec
6	21:17:18	SET autocommit = 0	0 row(s) affected	0.000 sec
7	21:17:18	START TRANSACTION	0 row(s) affected	0.016 sec
8	21:17:18	INSERT INTO CLASS VALUES (5,'Rahul')	Error Code: 1146. Table 'experiment4.class' doesn't exist	0.000 sec

Check current table-

SELECT * FROM CLASS;

Rollback to B (removes ID=7)-

ROLLBACK TO B;

SELECT * FROM CLASS;

Rollback to A (removes ID=6)-

ROLLBACK TO A;

SELECT * FROM CLASS;

Retrieval Queries (EMPLOYEES Table)-

1. Employee id, names, salaries--

SELECT Employee_Id, First_Name, Last_Name, Salary FROM EMPLOYEES;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Salary
101	Amit	Sharma	5000
102	Neha	Verma	4800
103	Ravi	Kumar	6000
104	Priya	Singh	4500
NULL	NULL	NULL	NULL

EMPLOYEES 2 | Apply | Revert

2. Employees under manager 100-

SELECT * FROM EMPLOYEES WHERE Manager_Id = 100;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	Amit	Sharma	amit@example.com	9999999999	2020-01-01	DEV	5000	0.10	100	60
103	Ravi	Kumar	ravi@example.com	7777777777	2019-05-20	MGR	6000	0.15	100	80
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

EMPLOYEES 3 | Apply | Revert

3. Salary >= 4800-

SELECT First_Name, Last_Name FROM EMPLOYEES WHERE Salary >= 4800;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

First_Name	Last_Name
Amit	Sharma
Neha	Verma
Ravi	Kumar

EMPLOYEES 4

4. Last name = 'AUSTIN'-

SELECT * FROM EMPLOYEES WHERE Last_Name = 'AUSTIN';

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

EMPLOYEES 5 | Apply | Revert

5. Departments 60, 70, 80-

SELECT * FROM EMPLOYEES WHERE Department_Id IN (60,70,80);

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	Amit	Sharma	amit@example.com	9999999999	2020-01-01	DEV	5000	0.10	100	60
102	Neha	Verma	neha@example.com	8888888888	2021-02-15	HR	4800	NULL	101	70
103	Ravi	Kumar	ravi@example.com	7777777777	2019-05-20	MGR	6000	0.15	100	80
104	Priya	Singh	priya@example.com	6666666666	2022-07-10	DEV	4500	NULL	102	60
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

EMPLOYEES 6 | Apply | Revert

6. Unique Manager IDs-

SELECT DISTINCT Manager_Id FROM EMPLOYEES;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Manager_Id
100
101
102

EMPLOYEES 7 × Read Only

Updating Records (Client12 Table)-

Clients with bal_due > 5000-

SELECT * FROM Client12 WHERE Bal_due > 5000;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ClientNO	Name	Address	Bal_due
C123	Ravi	Delhi	5100
C124	Neha	Mumbai	6000
C126	Priya	Kolkata	7000
NULL	NULL	NULL	NULL

Client12 8 × Apply Revert

Update balance for ClientNO C123-

UPDATE Client12 SET Bal_due = 5100 WHERE ClientNO = 'C123';

44	21:27:34	SELECT * FROM Client12 WHERE Bal_due > 5000 LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
45	21:28:06	UPDATE Client12 SET Bal_due = 5100 WHERE ClientNO = 'C123'	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec